

1. Prove $m(a+bX) = a + b \times m(X)$

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N (a + bx_i)$$

$$m(a+bX) = \frac{1}{N} \left[\sum_{i=1}^N a + \sum_{i=1}^N bx_i \right]$$

↑ ↑
constants

$$= \frac{1}{N} \left[Na + b \sum_{i=1}^N x_i \right]$$

$$m(a+bX) = a + b \underbrace{\frac{1}{N} \sum_{i=1}^N x_i}_{m(X)}$$

$$m(a+bX) = a + b m(X)$$

2. Show that $\text{cov}(X, a+bY) = b \text{cov}(X, Y)$

Using part 1, $m(a+bX) = a + b m(X)$ 

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) \left[\cancel{a} + by_i - (\cancel{a} + b m(Y)) \right]$$

$$\text{cov}(x, a+by) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) \cdot b(y_i - m(y))$$

$$\downarrow$$

$$= b \left[\frac{1}{N} \sum_{i=1}^N (x_i - m(x)) \cdot (y_i - m(y)) \right]$$

$$= b \cdot \text{cov}(x, y)$$

3a) Show that $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$

Reuse Q1 again: $m(a+bX) = a + b m(X)$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N \left[\cancel{(a+bX_i)} + \cancel{(a-bm(X))} \right]^2$$

$$= \frac{1}{N} \sum_{i=1}^N b^2 \underbrace{[X_i - m(X)]^2}_{= \text{cov}(X, X)}$$

$$\therefore \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$$

3b) Given $s^2 = \frac{1}{N} \sum_{i=1}^N [(X_i - m(X))]^2 = \text{cov}(X, X),$

$$s^2 = \text{cov}(X, X)$$

4. I don't think so, but let's check

Let $g(x) = e^x$ and dataset $X = [4, 6]$, $N=2$
 $m=5$

$$g(m(x)) = e^5$$

$$g(x) = [e^4, e^6] \quad m(g(x)) = \frac{e^4 + e^6}{2}$$
$$= \frac{e^4(1 + e^2)}{2}$$

$$\frac{1 + e^2}{2} \neq e,$$

$\therefore$ they are
not equal.